

MA(3)

$$Y_t = e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3}$$

$$e_t \stackrel{\text{isi}}{\sim} (0, \sigma^2)$$

$$\begin{aligned} \Rightarrow E(Y_t) &= E(e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3}) \\ &= E(e_t) - \theta_1 E(e_{t-1}) - \theta_2 E(e_{t-2}) - \theta_3 E(e_{t-3}) \\ &= 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow \gamma_0 = \text{Var}(Y_t) &= \text{Var}(e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3}) \\ &= \text{Var}(e_t) - \text{Var}(\theta_1 e_{t-1}) - \text{Var}(\theta_2 e_{t-2}) - \text{Var}(\theta_3 e_{t-3}) \\ &= \sigma_e^2 - \theta_1^2 \sigma_e^2 - \theta_2^2 \sigma_e^2 - \theta_3^2 \sigma_e^2 \\ &= (1 - \theta_1^2 - \theta_2^2 - \theta_3^2) \sigma_e^2 \end{aligned}$$

$$\begin{aligned} \Rightarrow \gamma_1 = \text{Cov}(Y_t, Y_{t-1}) &= \text{Cov}(e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3}, e_{t-1} - \theta_1 e_{t-2} - \theta_2 e_{t-3} - \theta_3 e_{t-4}) \\ &= \text{Cov}(-\theta_1 e_{t-1}, e_{t-1}) + \text{Cov}(-\theta_2 e_{t-2}, -\theta_1 e_{t-2}) + \text{Cov}(-\theta_3 e_{t-3}, -\theta_2 e_{t-3}) \\ &= (-\theta_1 + (-\theta_2)(-\theta_1) + (-\theta_3)(-\theta_2)) \sigma_e^2 \\ &= (-\theta_1 + \theta_1 \theta_2 + \theta_2 \theta_3) \sigma_e^2 // \end{aligned}$$

$$\begin{aligned} \Rightarrow \gamma_2 = \text{Cov}(Y_t, Y_{t-2}) &= \text{Cov}(e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3}, e_{t-2} - \theta_1 e_{t-3} - \theta_2 e_{t-4} - \theta_3 e_{t-5}) \\ &= \text{Cov}(-\theta_2 e_{t-2}, e_{t-2}) + \text{Cov}(-\theta_3 e_{t-3}, -\theta_1 e_{t-3}) \\ &= (-\theta_2 + (-\theta_3)(-\theta_1)) \sigma_e^2 \\ &= (-\theta_2 + \theta_1 \theta_3) \sigma_e^2 // \end{aligned}$$

$$\begin{aligned} \Rightarrow \gamma_3 = \text{Cov}(Y_t, Y_{t-3}) &= \text{Cov}(e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \theta_3 e_{t-3}, e_{t-3} - \theta_1 e_{t-4} - \theta_2 e_{t-5} - \theta_3 e_{t-6}) \\ &= \text{Cov}(-\theta_3 e_{t-3}, e_{t-3}) \\ &= -\theta_3 \sigma_e^2 // \end{aligned}$$

$$\Rightarrow \rho_1 = \frac{\text{Cov}(Y_t, Y_{t-1})}{\sqrt{\text{Var}(Y_t) \text{Var}(Y_{t-1})}} = \frac{(-\theta_1 + \theta_1 \theta_2 + \theta_2 \theta_3) \sigma_e^2}{(1 - \theta_1^2 - \theta_2^2 - \theta_3^2) \sigma_e^2} = \frac{-\theta_1 + \theta_1 \theta_2 + \theta_2 \theta_3}{1 - \theta_1^2 - \theta_2^2 - \theta_3^2}$$

$$\rho_2 = \frac{\text{Cov}(Y_t, Y_{t-2})}{\sqrt{\text{Var}(Y_t) \text{Var}(Y_{t-2})}} = \frac{(-\theta_2 + \theta_1 \theta_3) \sigma_e^2}{(1 - \theta_1^2 - \theta_2^2 - \theta_3^2) \sigma_e^2} = \frac{-\theta_2 + \theta_1 \theta_3}{1 - \theta_1^2 - \theta_2^2 - \theta_3^2}$$

$$\rho_2 = \frac{\text{Cov}(Y_t, Y_{t-2})}{\sqrt{\text{Var}(Y_t)\text{Var}(Y_{t-2})}} = \frac{(-\theta_2 + \theta_1\theta_3)\sigma_e^2}{(1-\theta_1^2-\theta_2^2-\theta_3^2)\sigma_e^2} = \frac{-\theta_2 + \theta_1\theta_3}{1-\theta_1^2-\theta_2^2-\theta_3^2}$$

$$\rho_3 = \frac{\text{Cov}(Y_t, Y_{t-3})}{\sqrt{\text{Var}(Y_t)\text{Var}(Y_{t-3})}} = \frac{-\theta_3\sigma_e^2}{(1-\theta_1^2-\theta_2^2-\theta_3^2)\sigma_e^2} = \frac{-\theta_3}{1-\theta_1^2-\theta_2^2-\theta_3^2}$$

$$\rho_k = 0, \text{ untuk } k = 4, 5, \dots$$

AR(3)

$$Y_t = e_t + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \phi_3 Y_{t-3}$$

$$e_t \stackrel{\text{bsi}}{\sim} (0, \sigma^2)$$